

What is AI?

Christoph Benz Müller
U Bamberg (& FU Berlin)

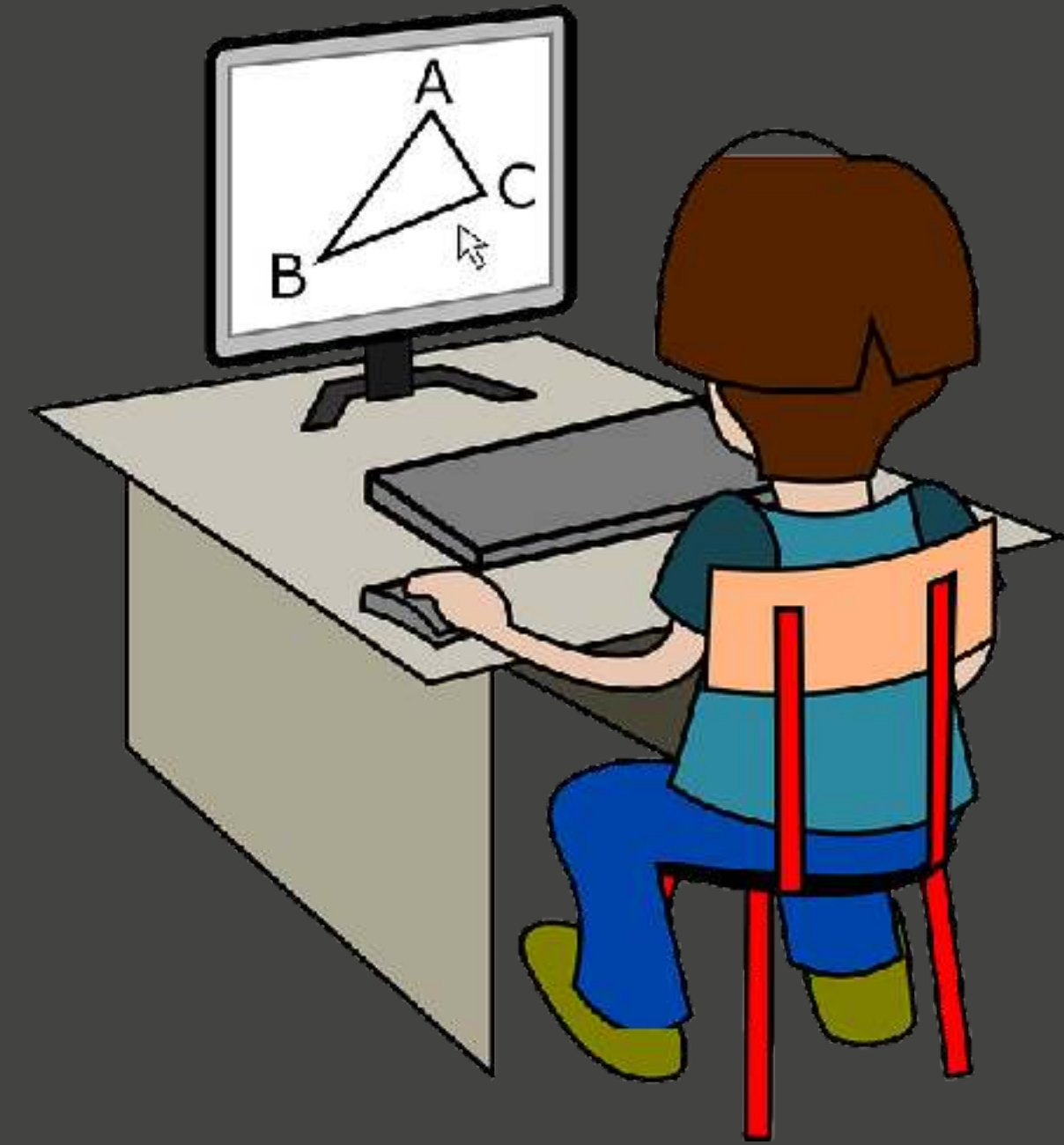


Summer School on AI & Ethics, Bamberg, September 26-30, 2022

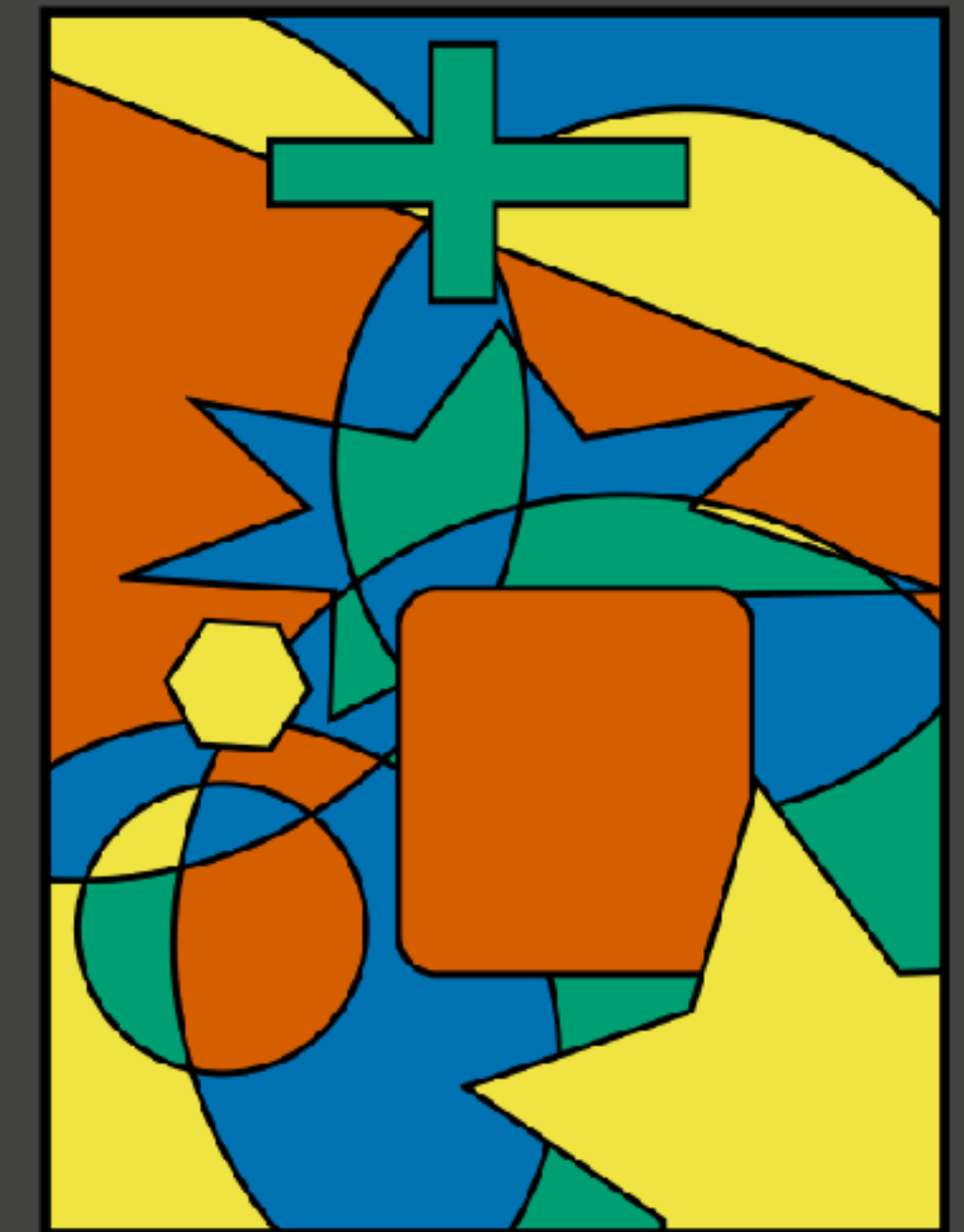
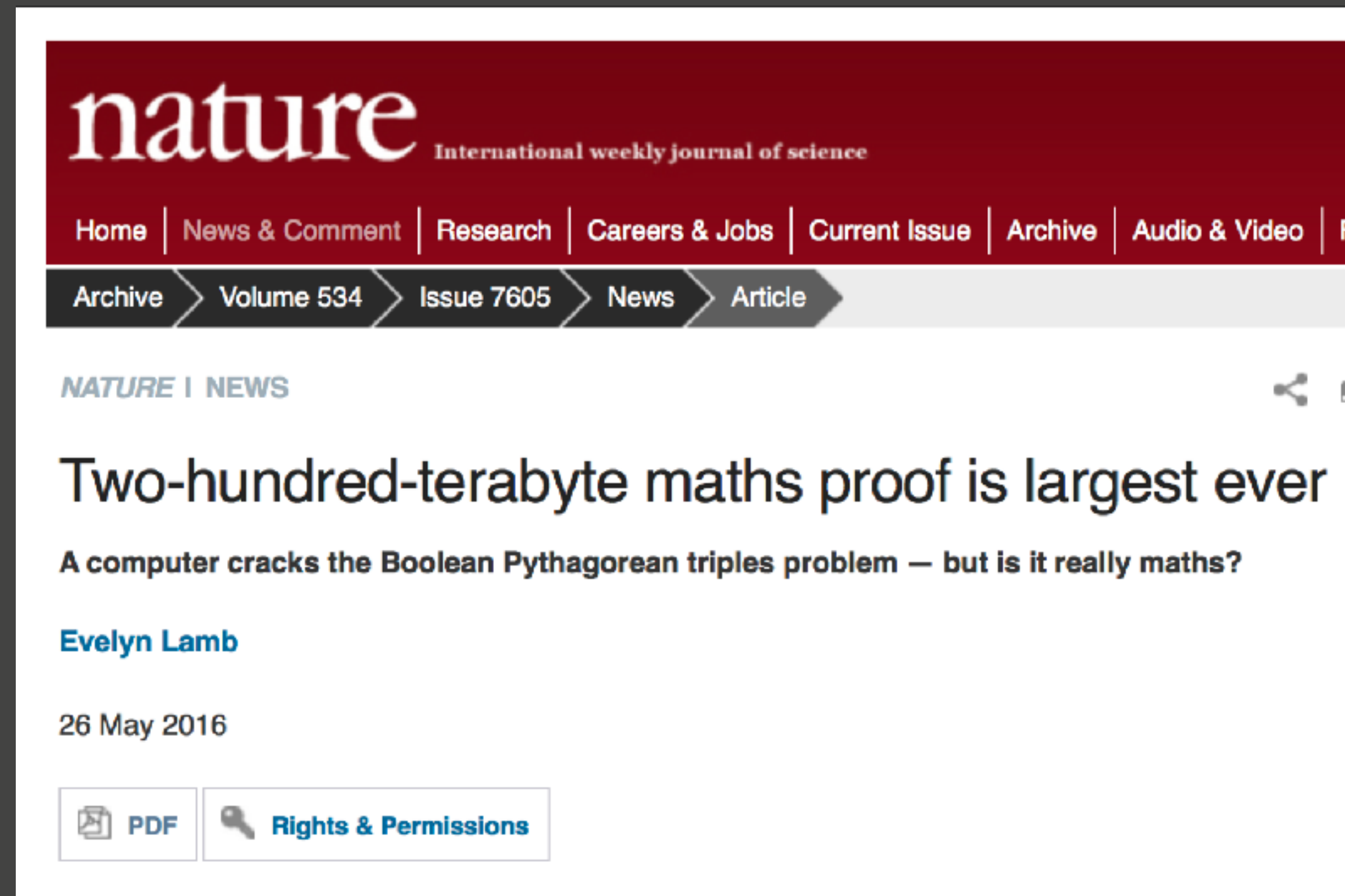
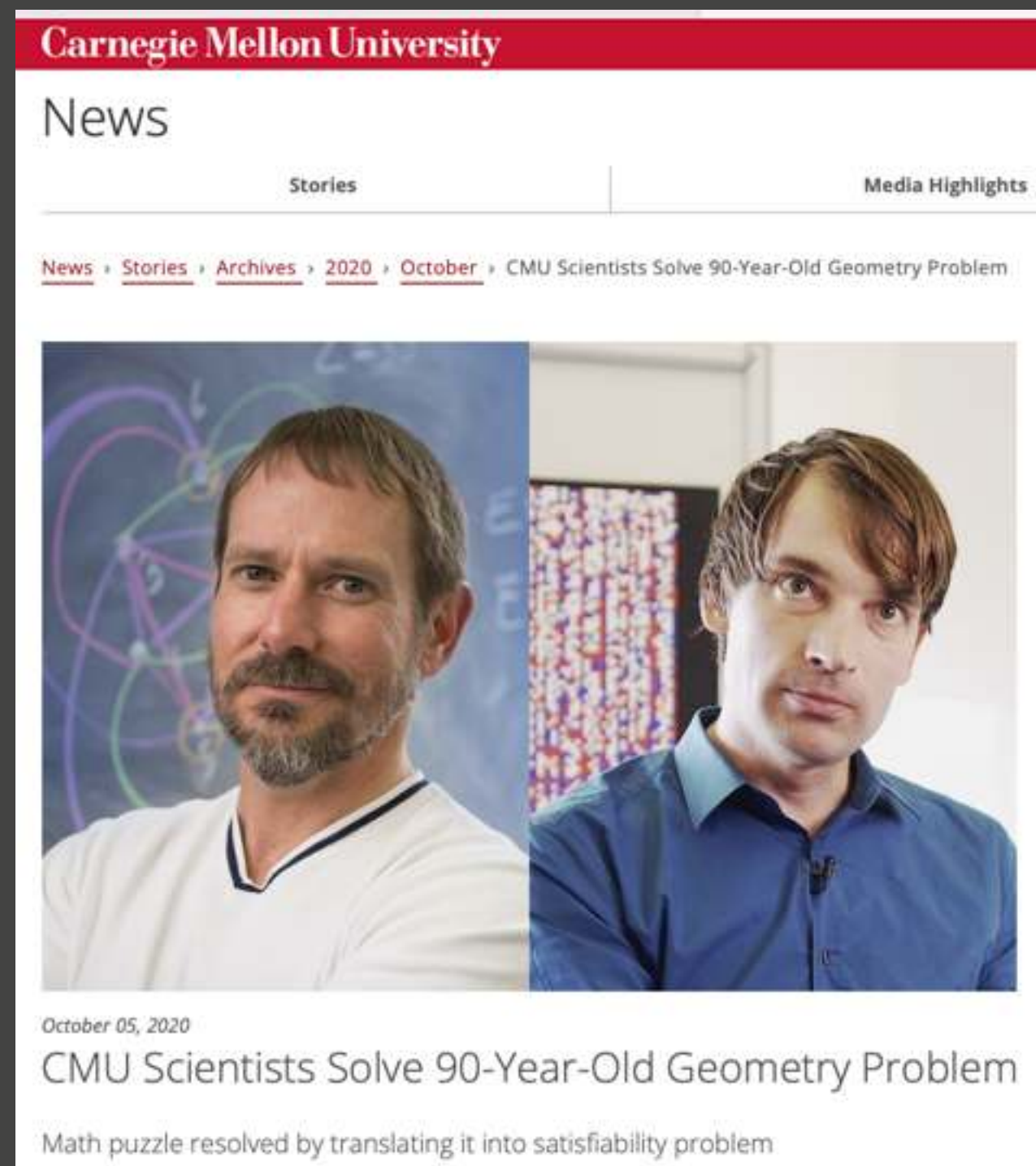
PRÄLUDIUM: QUIZ-QUESTION?

CAN AI SYSTEMS SOLVE OPEN
MATHEMATICAL PROBLEMS?

WHICH AI SYSTEMS?



PRÄLUDIUM: QUIZ-QUESTION?



Four color theorem solved with theorem provers already in 1976

SAT-Solving and Theorem Proving: original subfields of AI
SAT-solving very successful, got embraced by general CS

TAKE AWAY: THERE IS AI SUCCESS BEYOND ML

HOW TO START SUCH A TALK?

LOVE

PASSION

EMOTION

I felt in love with the field of AI about 30 years ago because of a provocative statement by Jörg Siekmann in his AI lecture course

And for the study of strong AI
these topics are of particular interest

AI & LOVE

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WHERE LOVE FALLS ...

... until recently:

... in today's online dating:

Destiny

determined by **AI-algorithms**



Swan "Petra" falls hopelessly in love with a pedal boat (2006)

AI & LOVE

Peter and Katharina
in love with "Saugi"



Pete und Cathy
in love with "Dolly"

WHAT IS AI?

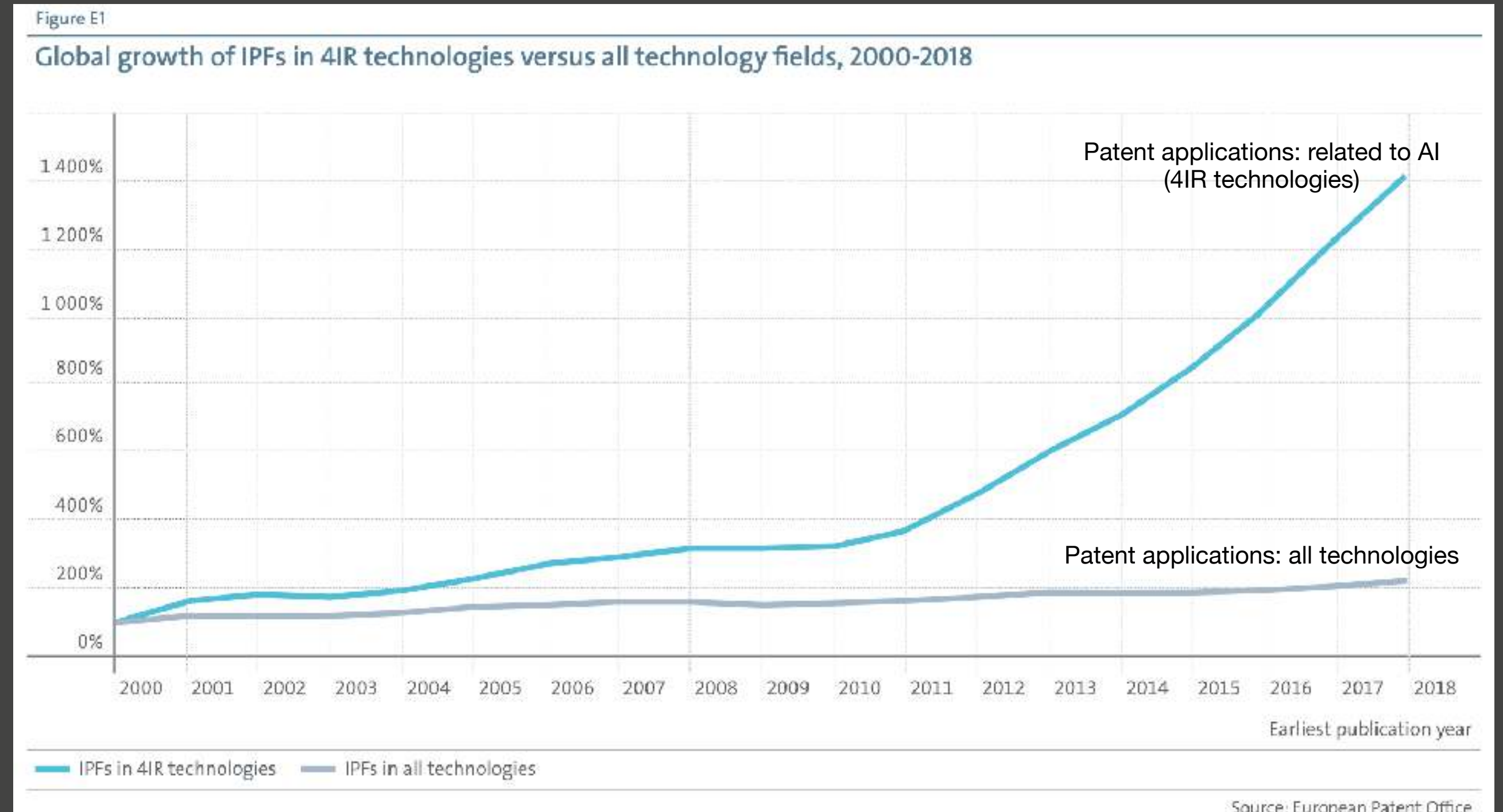
A huge market

Warum Deutschland mehr in KI investieren sollte

Künstliche Intelligenz ist für Wirtschaft folgenreicher als die Dampfmaschine

Von **Mirjam Hecking**
07.09.2018, 12.07 Uhr

manager magazin



Source: Patents and the Fourth Industrial Revolution, European Patent Office, Dez. 2020

The steam engine of the 21st century?

WHAT IS AI?

AUTOMATISIERUNG

Millionen Arbeitnehmer müssen umschulen

VON JANNIK WAIDNER - AKTUALISIERT AM 19.02.2021 - 09:04

Frankfurter Allgemeine
ZEITUNG ● FAZ.NET

Source of social tensions and problems

WHAT IS AI?

Great help in solving
global environmental problems?

How Innovative AI Solutions Can Help Combat Global Warming

Oct 14, 2020, 10:00am EDT



Asokan Ashok Forbes Councils Member

Forbes Technology Council COUNCIL POST | Paid Program

Innovation

Artificial intelligence / Machine learning

Training a single AI model can emit as much carbon as five cars in their lifetimes

Deep learning has a terrible carbon footprint.

by **Karen Hao**

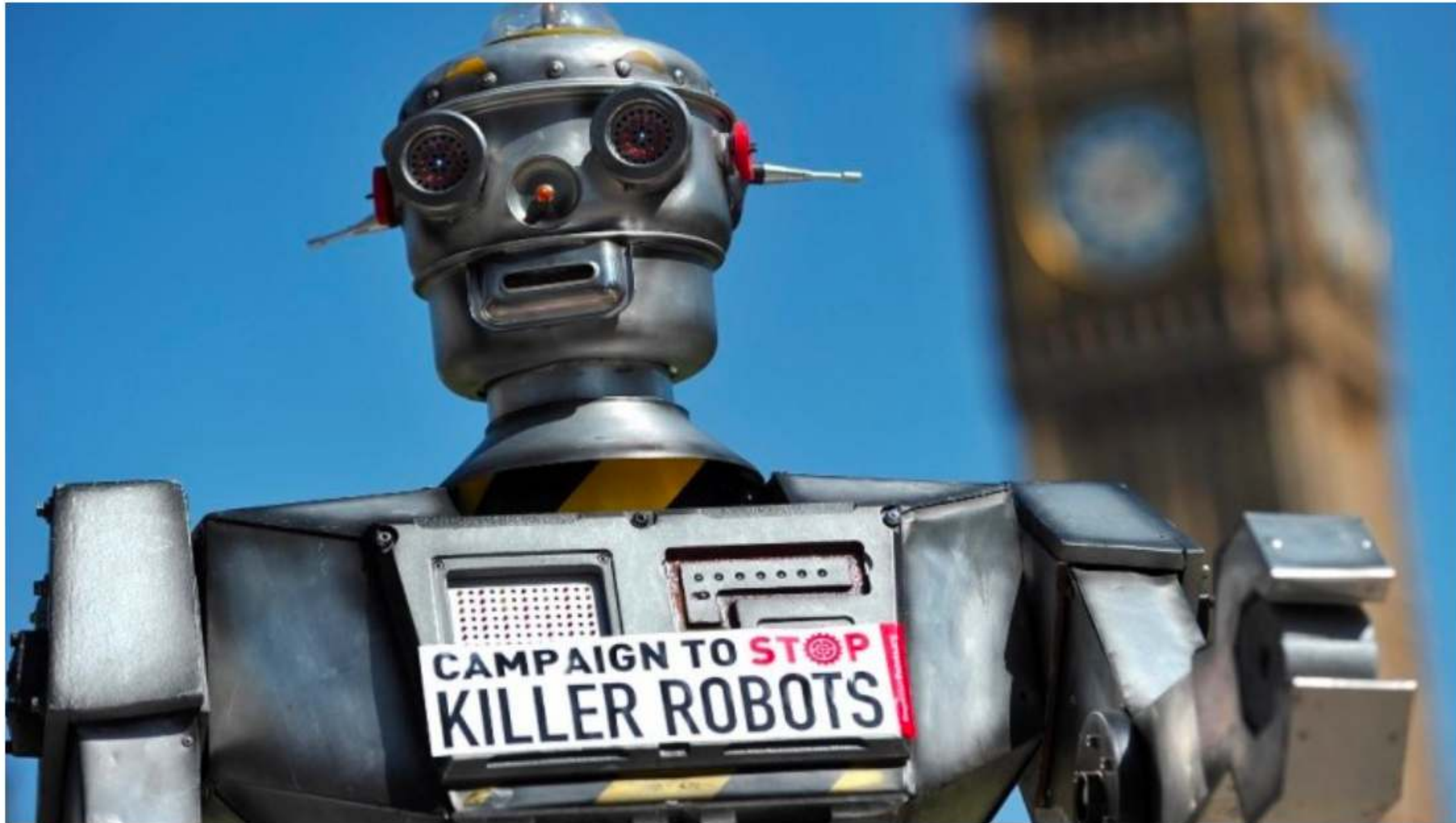
MIT Technology Review

June 6, 2019

Massive use of data-driven AI
could itself become a climate killer?

WHAT IS AI?

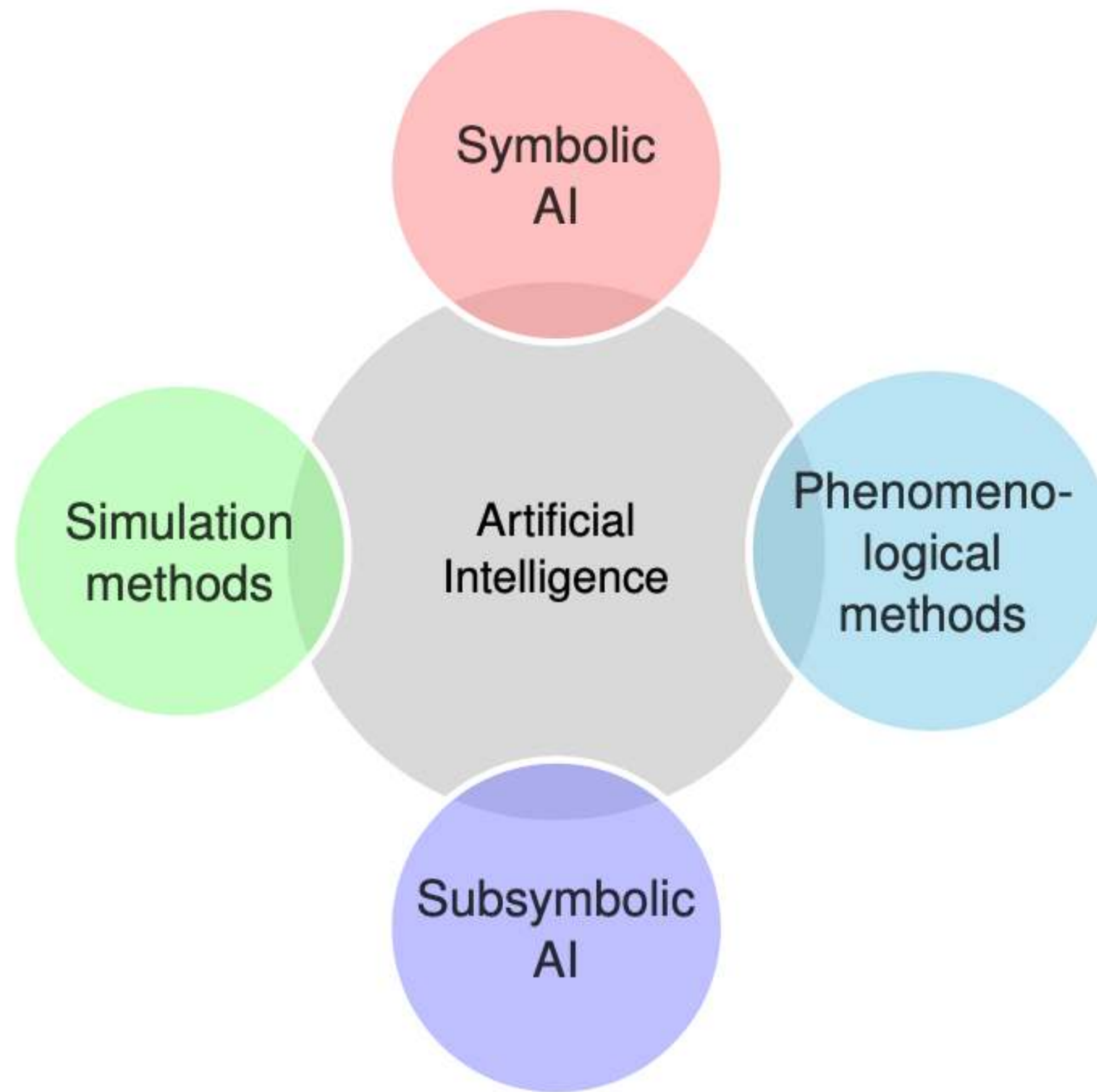
A DUAL USE TECHNOLOGY!



Source: Deutschlandfunk/AFP, Carl Court, 2019

A danger regarding military tension and escalation

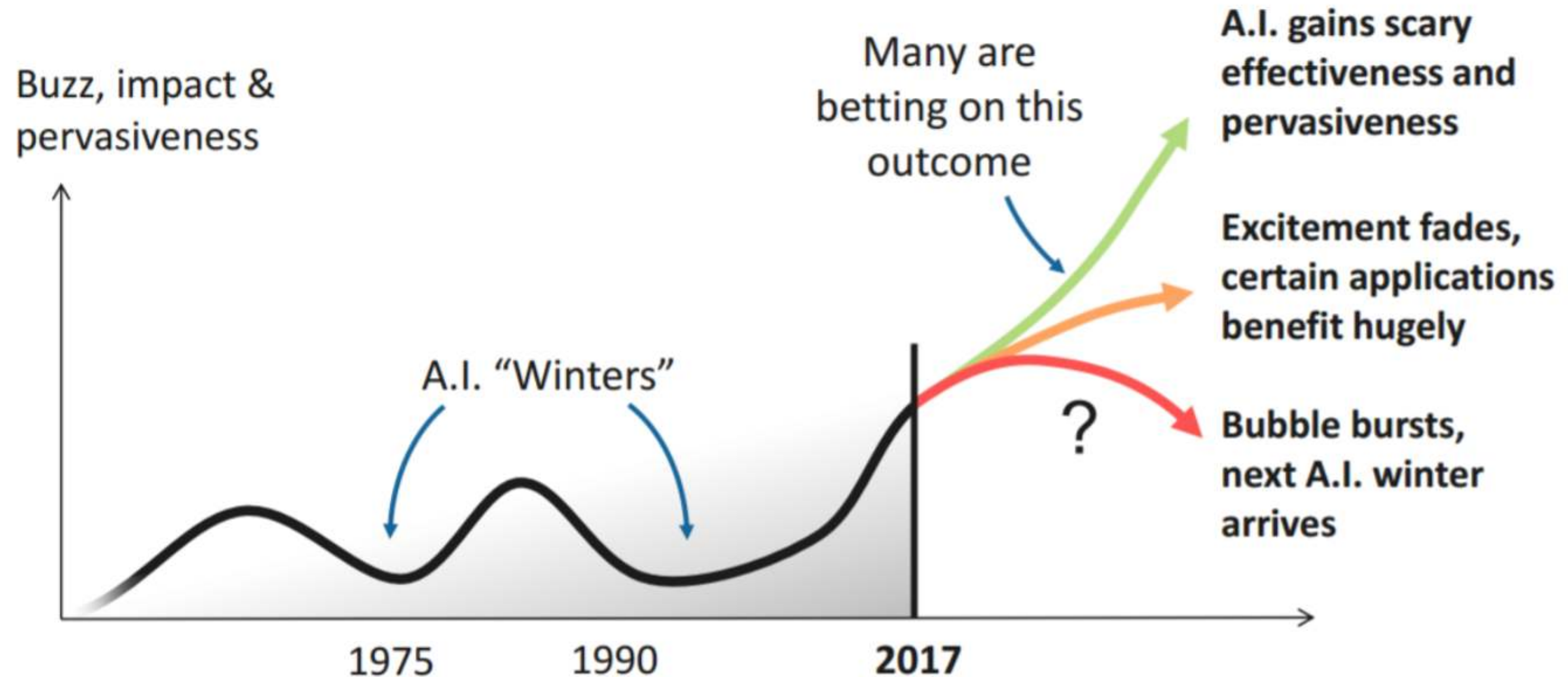
WHAT IS AI?



- ▶ Knowledge Representation
- ▶ Logical Reasoning
- ▶ Planning
- ▶ Multi-Agent Systems
- ▶ Search
- ▶ ...
- ▶ Robotics
- ▶ Pattern Recognition & Machine Learning
- ▶ Neural Networks

A science with various subfields

WHAT IS AI?

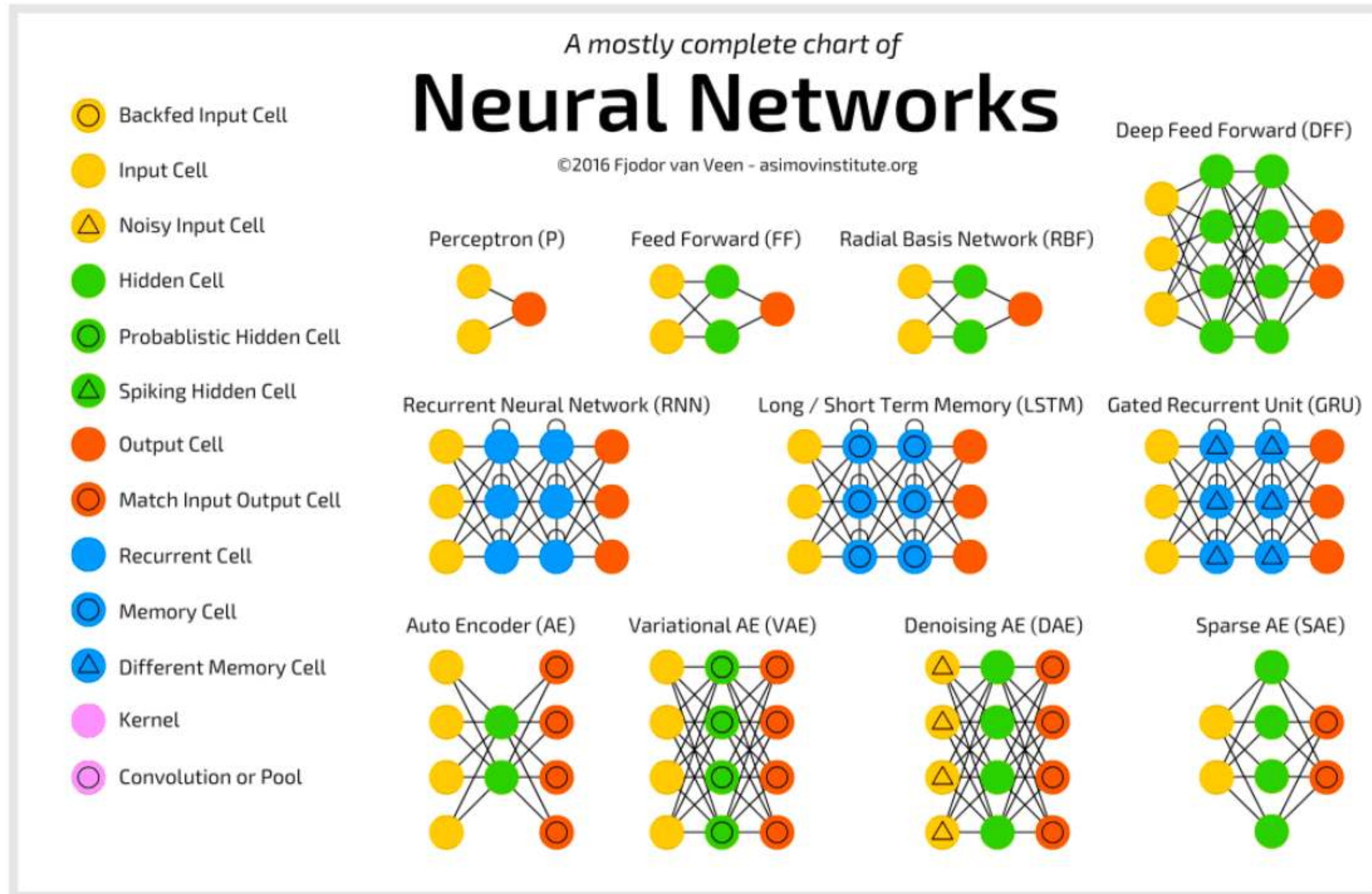


Source: [Quora thread on AI hype and winters](#)

A story of summers and winters — currently: summer

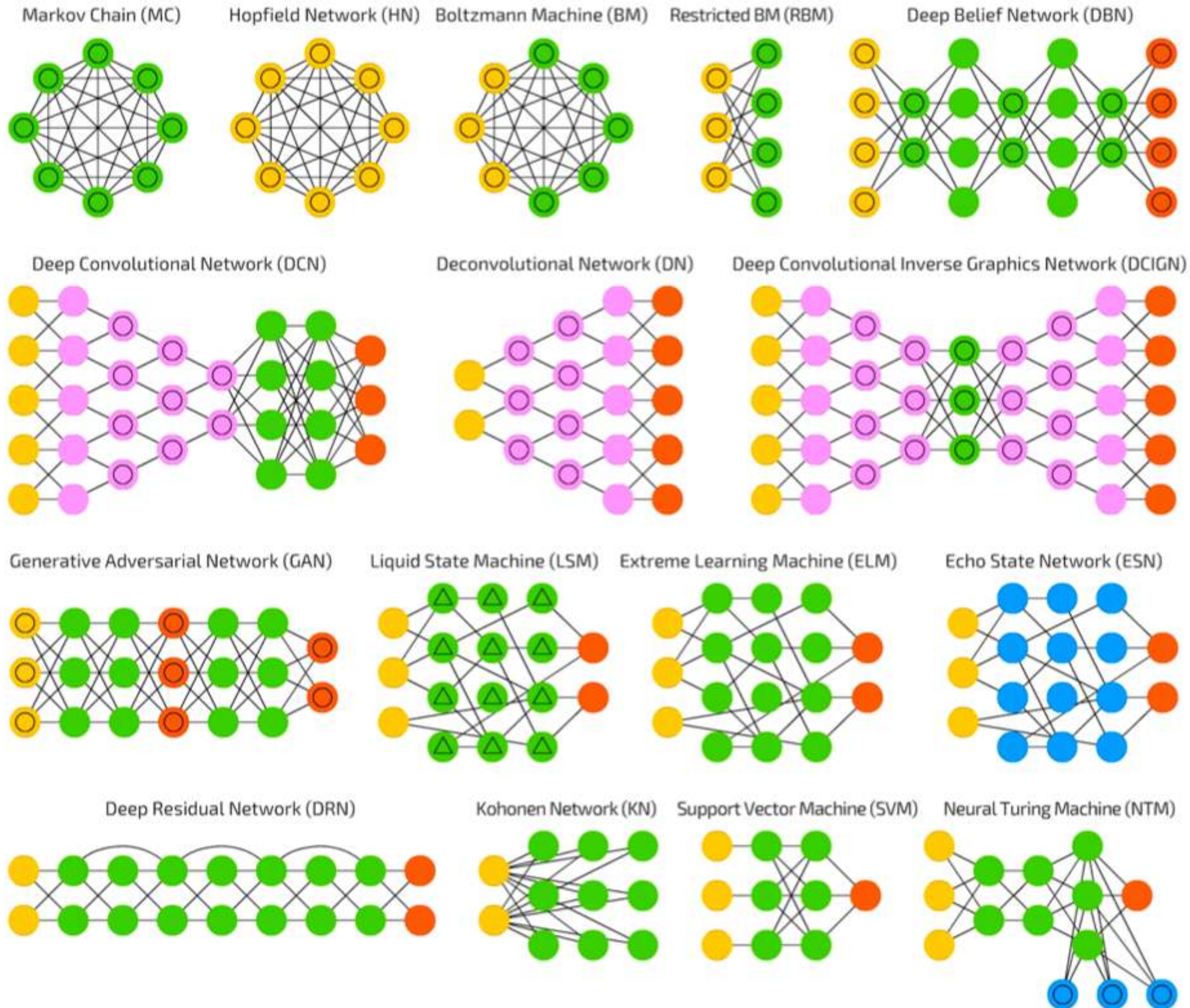
Neural Networks & Machine Learning

Source: Fjodor van Veen, asimovinstitute.org, 2016



Neural Networks & Machine Learning

Source: Fjodor van Veen, asimovinstitute.org, 2016



FF)

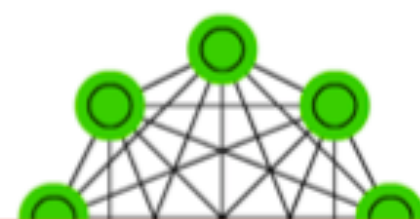
GRU)

(E)

Neural Networks & Machine Learning

Source: Fjodor van Veen, asimovinstitute.org, 2016

Markov Chain (MC)



Hopfield Network (HN)



Boltzmann Machine (BM)



Restricted BM (RBM)



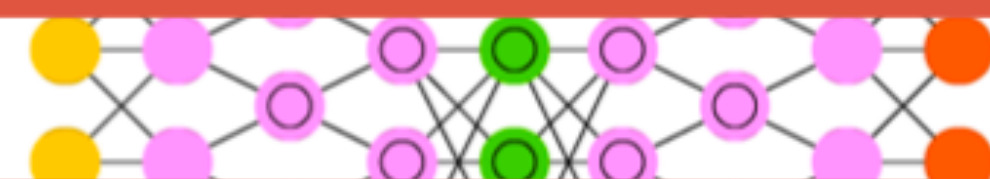
Deep Belief Network (DBN)



(FF)

Sub-/Non-Symbolic AI, since

- ▶ nodes do not carry any semantic meaning



(GRU)

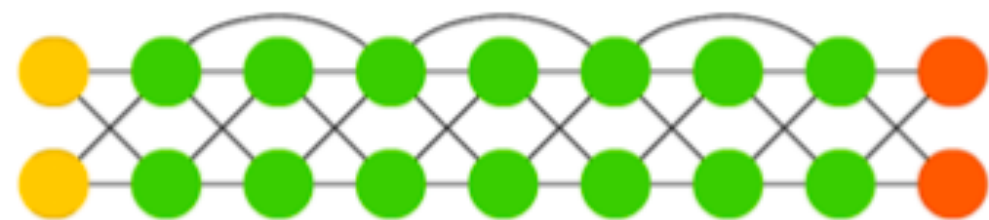
Pros:

- ▶ robust & very strong in specific domains

Cons:

- ▶ data-intensive training, requires substantial human expertise, intransparent, bias, adversarial attacks

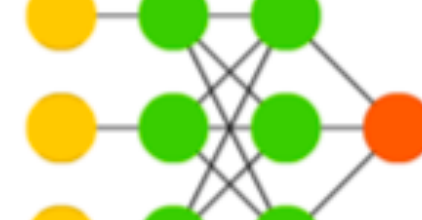
Deep Residual Network (DNN)



Convolutional Network (CNN)



Support Vector Machine (SVM)



Neural Turing Machine (NTM)



Symbolic AI – Example: Logical Reasoning

$A \cup B ::= \dots$

$A \cap B ::= \dots$

$A \subseteq B \Leftrightarrow \dots$

$A = B ::= \dots$

$\dots$

$\dots$

Logic Rules

Assumptions

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Theorem

Symbolic AI – Example: Logical Reasoning

$A \cup B ::= \dots$

$A \cap B ::= \dots$

$A \subseteq B \Leftrightarrow \dots$

$A = B ::= \dots$

$\dots$

$\dots$

Logic Rules

Assumptions

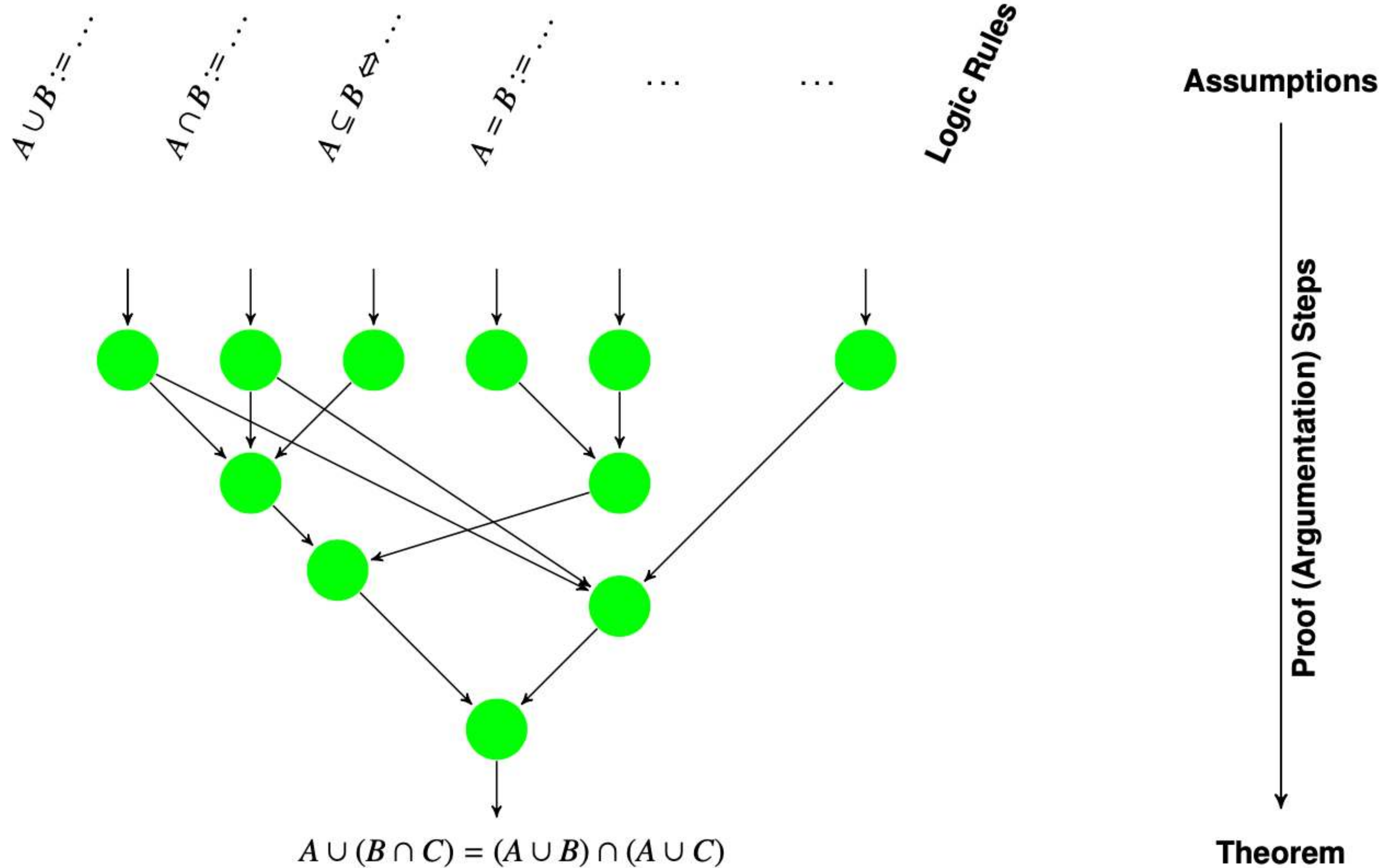
Proof (Argumentation) Steps



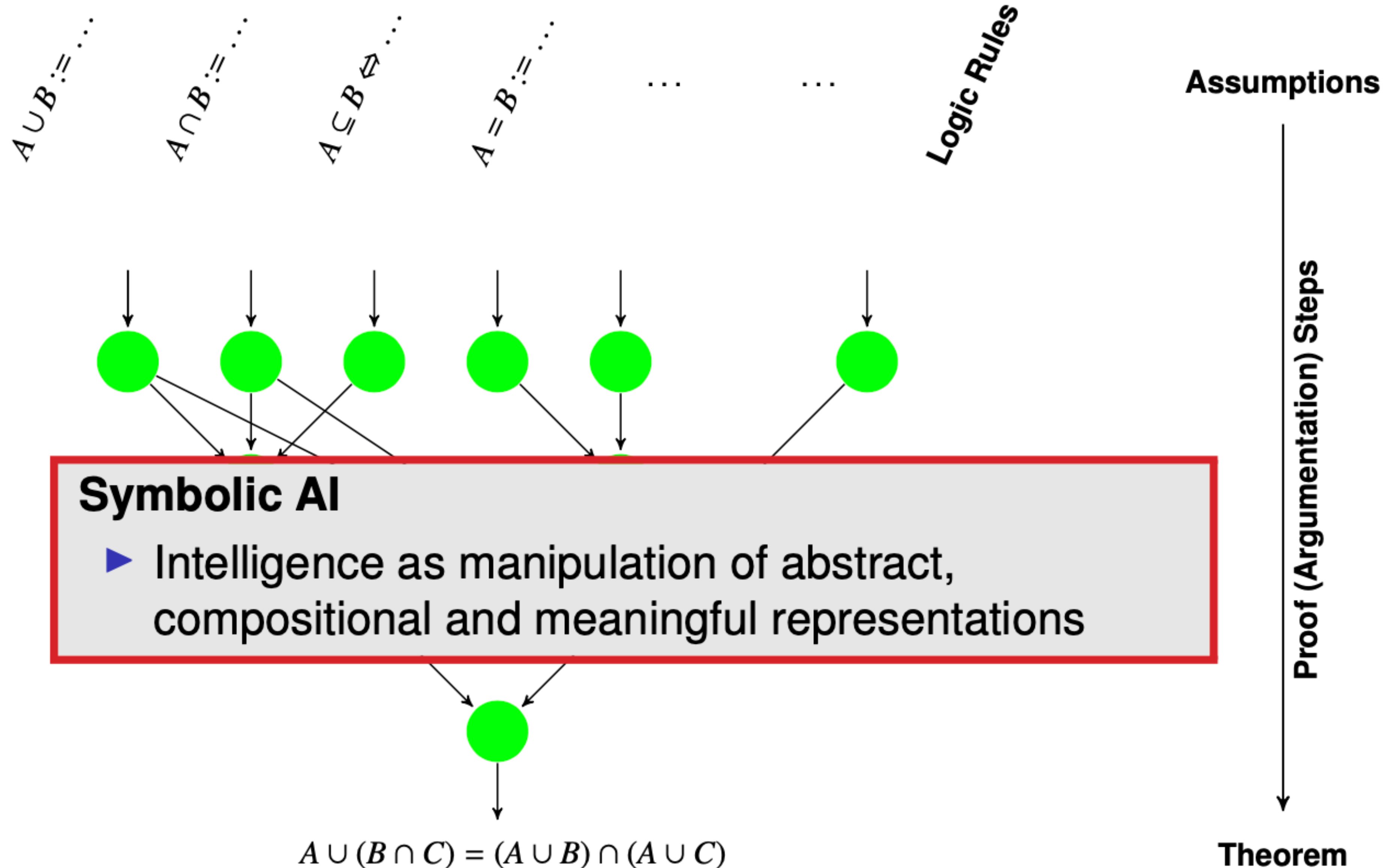
Theorem

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Symbolic AI – Example: Logical Reasoning



Symbolic AI – Example: Logical Reasoning

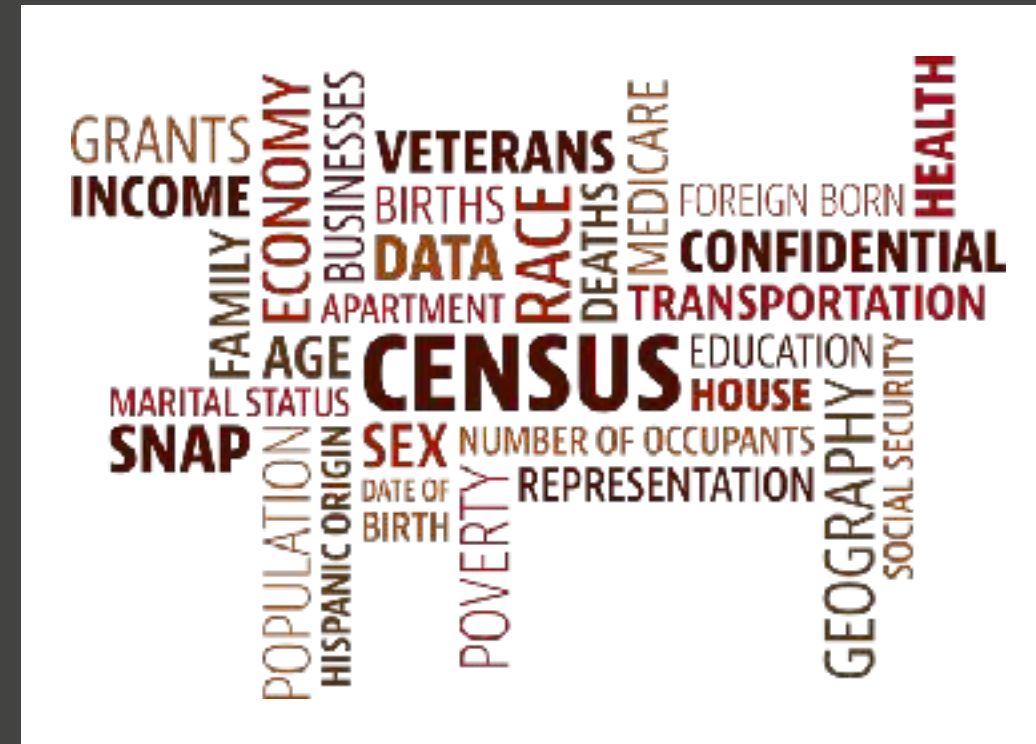


WHAT IS AI?

Machine Learning (ML) Hype

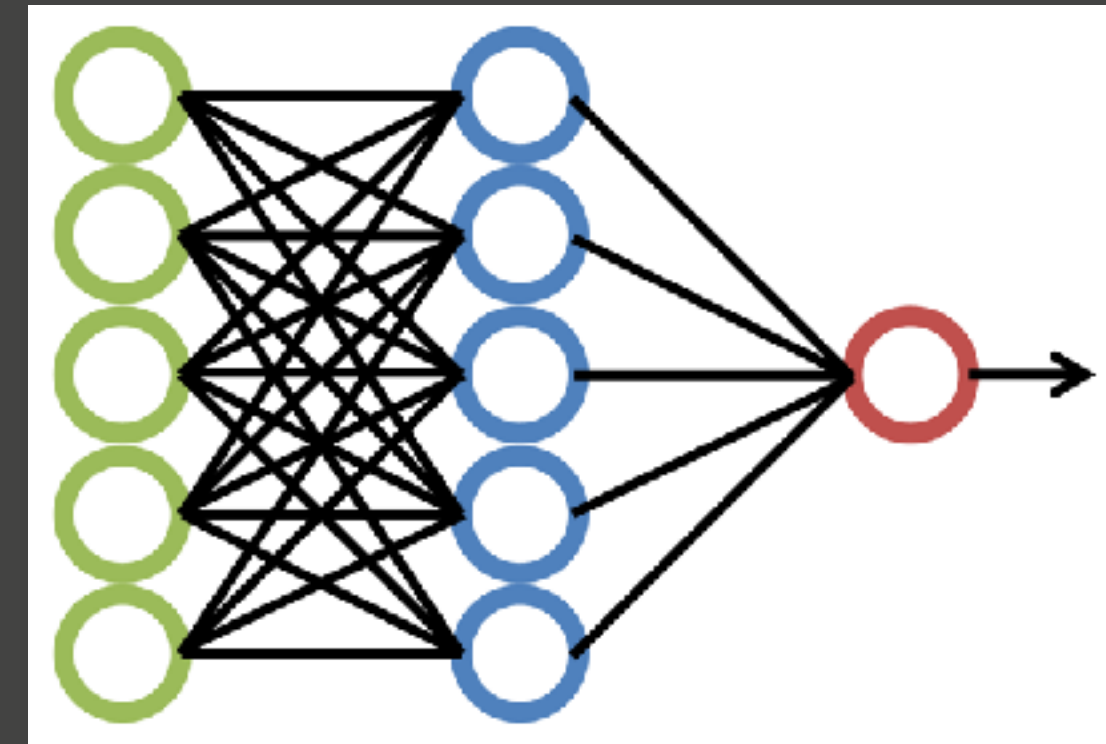
Many currently believe:

AI =



Data

+



ML/Deep Learning

Aha ... really?

WHAT IS AI?

Weak AI: ... solve specific problems

Strong AI: ... everything humans can do (and possibly way more)

Strong AI requires at least (own working hypothesis):

1. Problem Solving (in the sense of weak AI)
2. Exploring and Acting in Unknown Territory (e.g. Mars-Rover)
3. Abstract & Rational Reasoning (e.g. exploration of a new theory in maths)
4. Self-Reflection (e.g. detect own mistakes, questioning the results of one's own thinking)
5. Social Interaction (e.g. adjusting personal goals and values to those of a community)

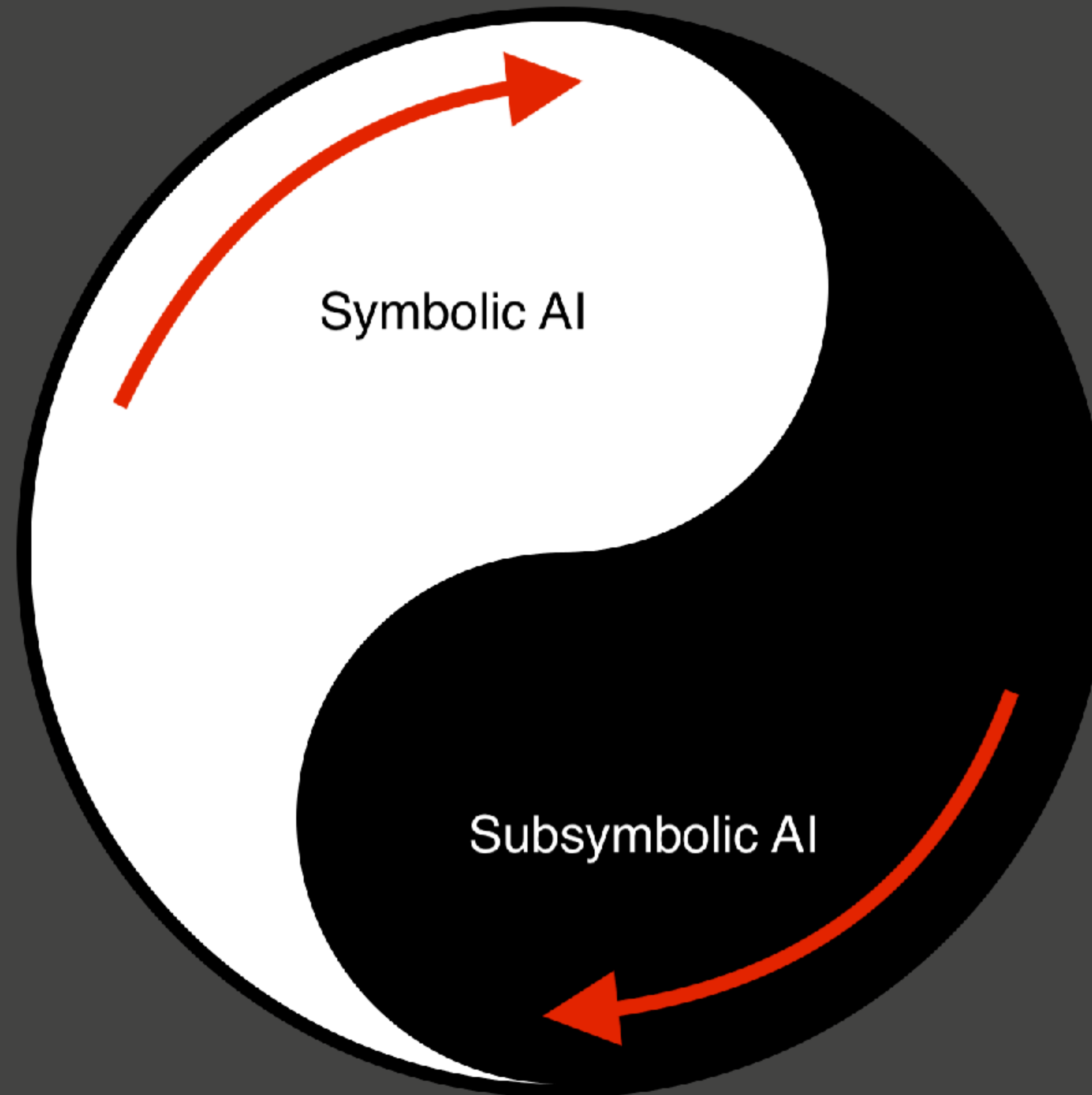
Machine Learning

Hybrid AI

Artificial Intelligence

symbolic

Mind
Reasoning
Deductive
Little Data
Causalities
Precise



subsymbolic

Brain
Learning
Inductive
Massive Data
Correlations
Robust

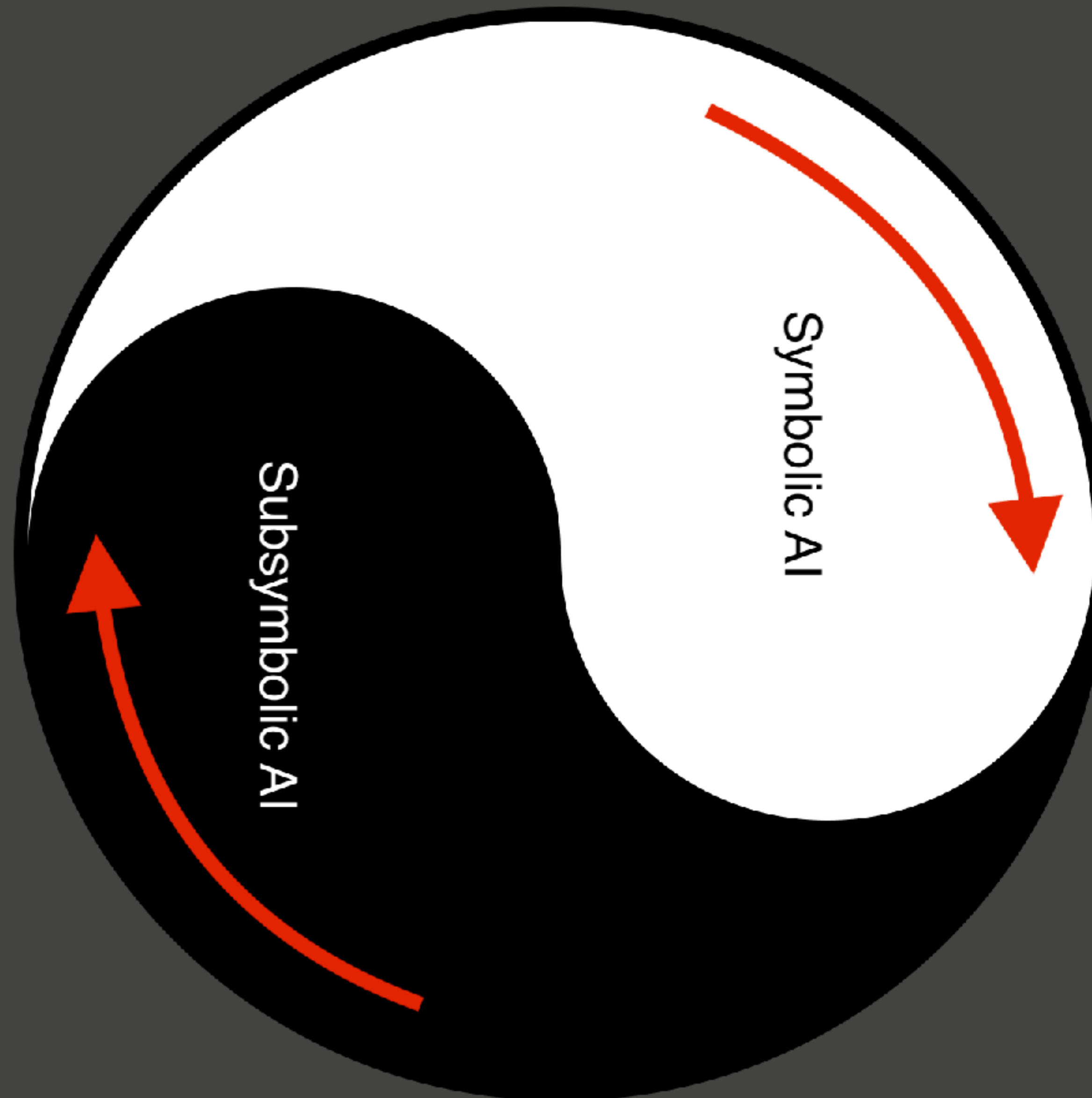
Artificial Intelligence

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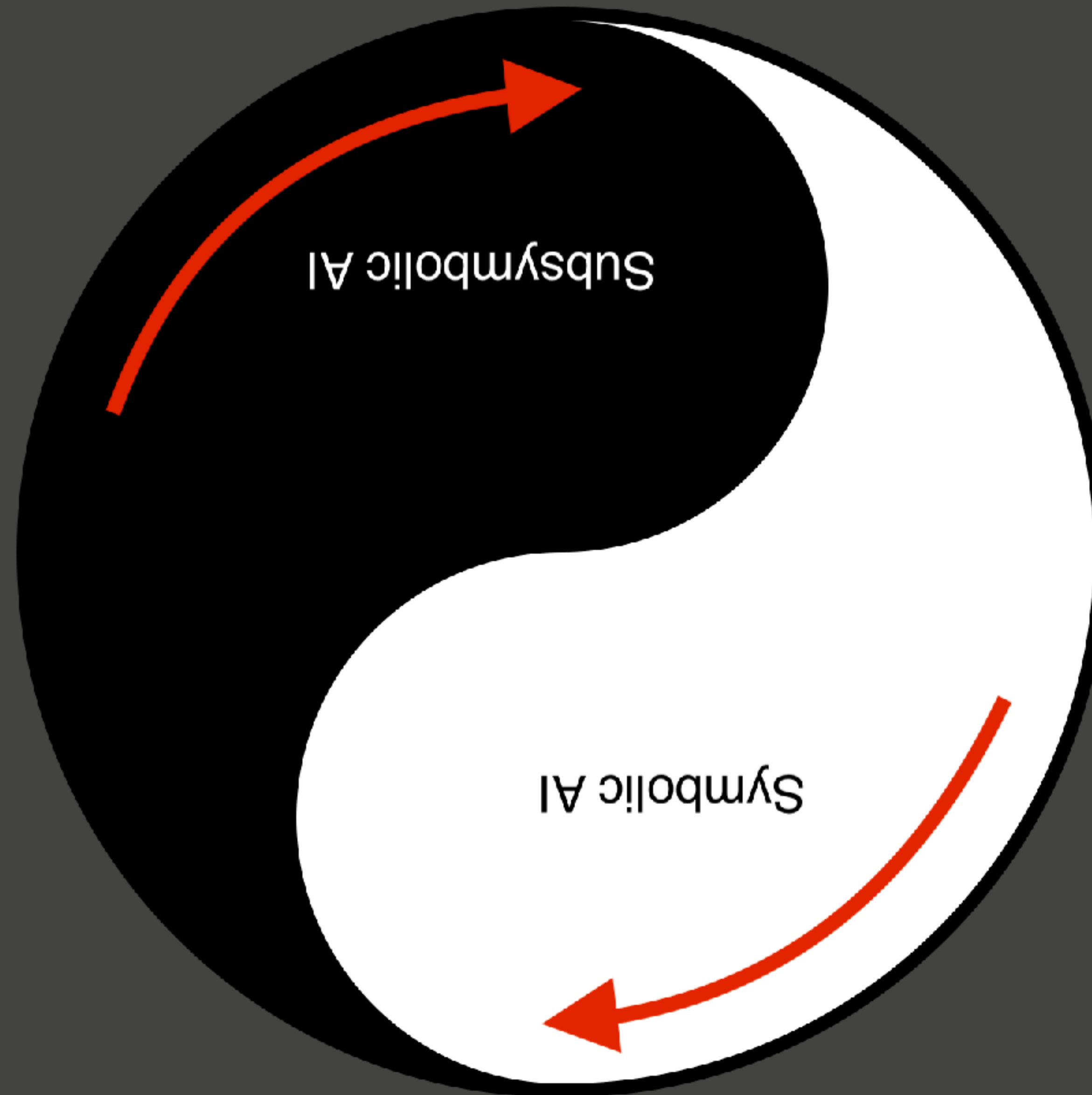
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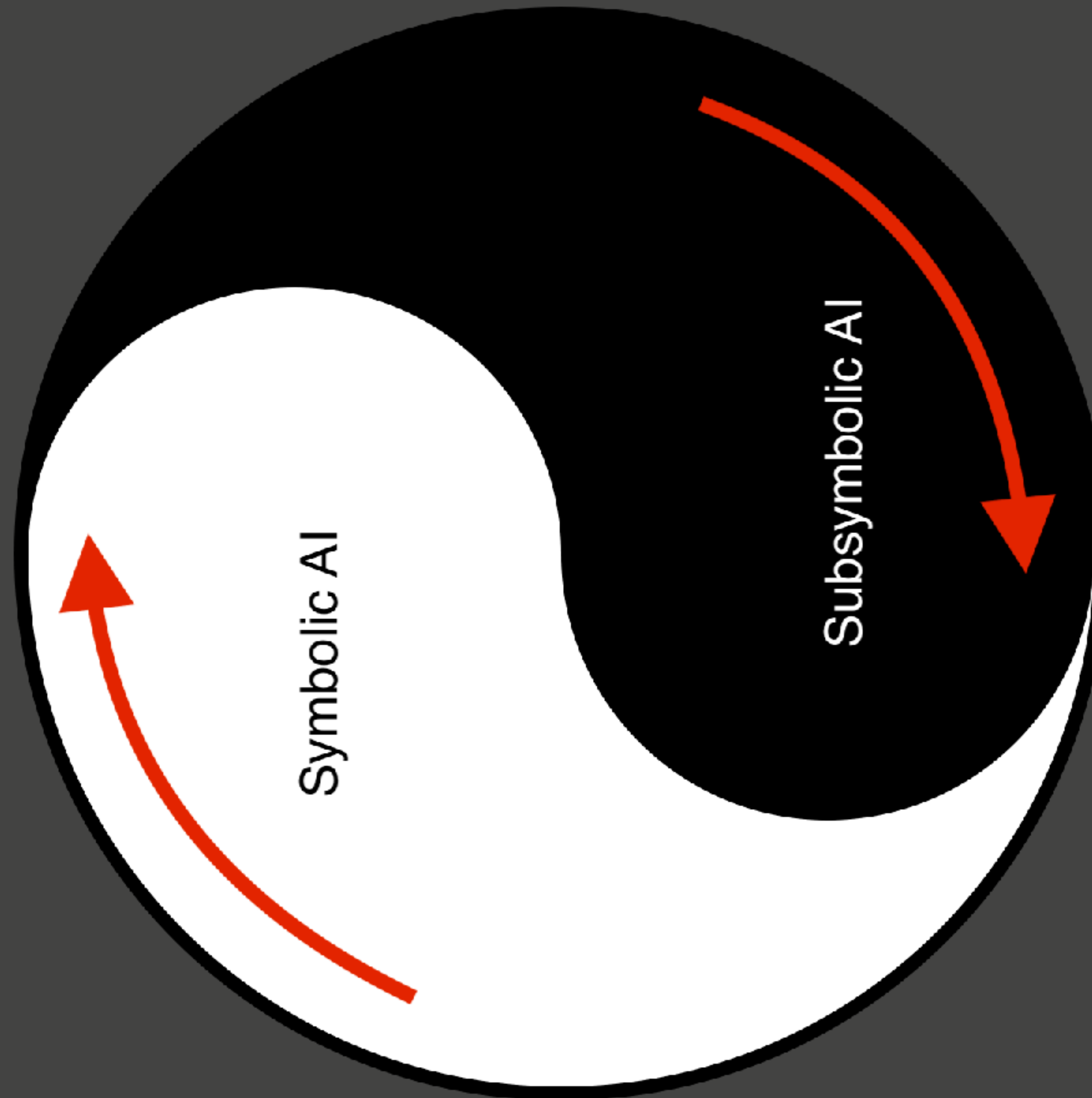
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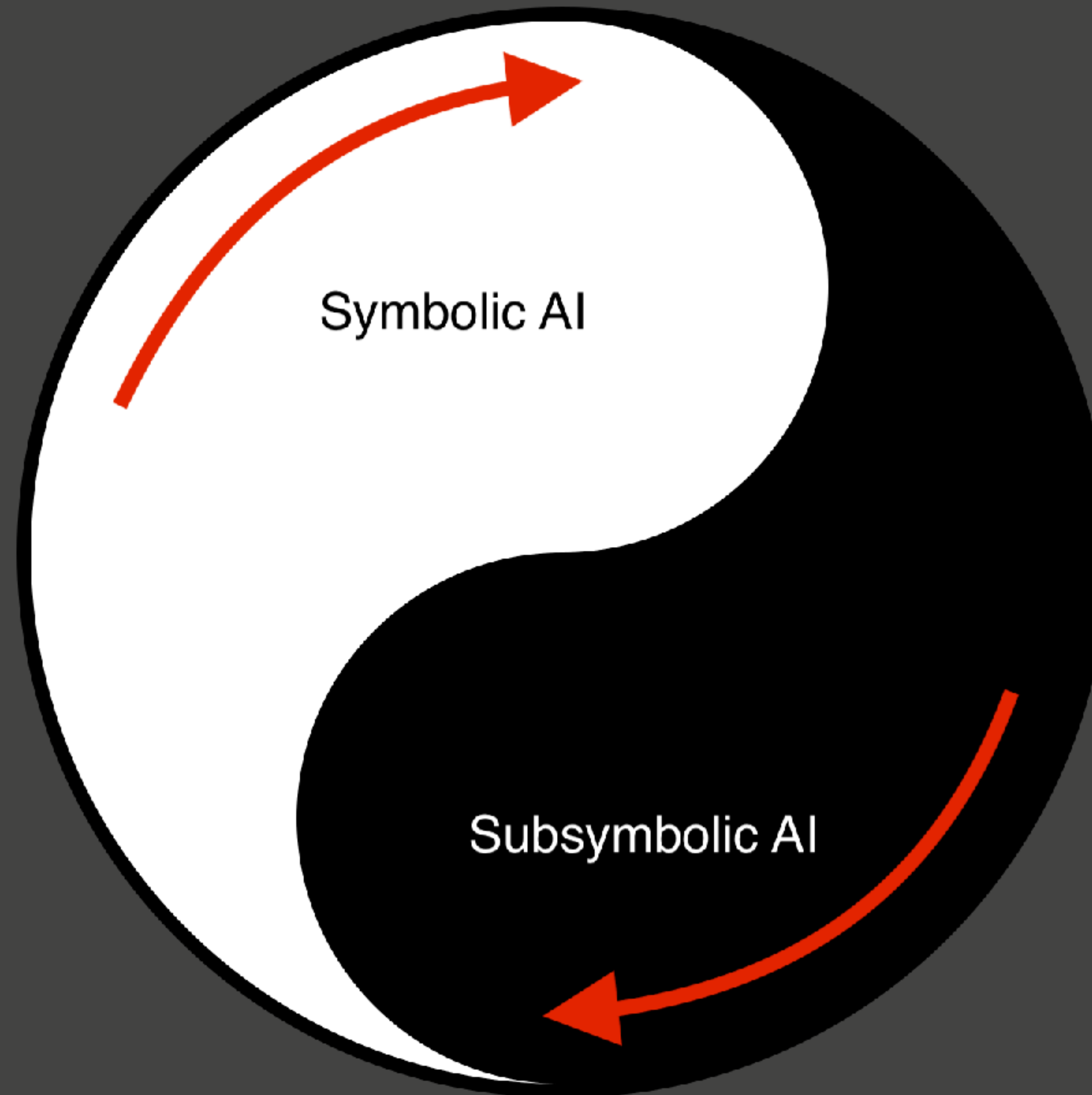
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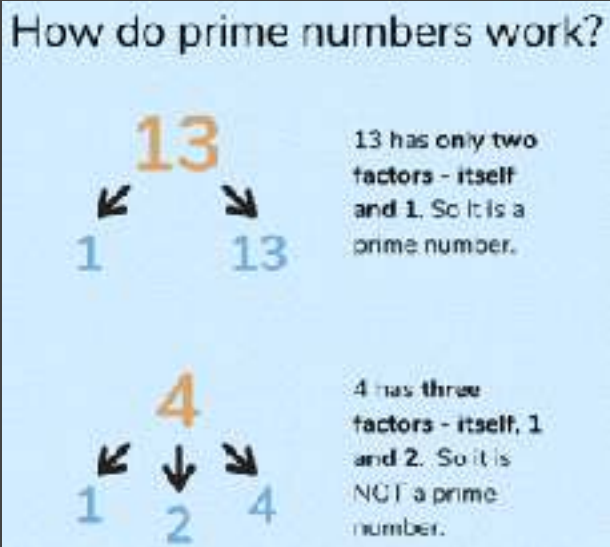
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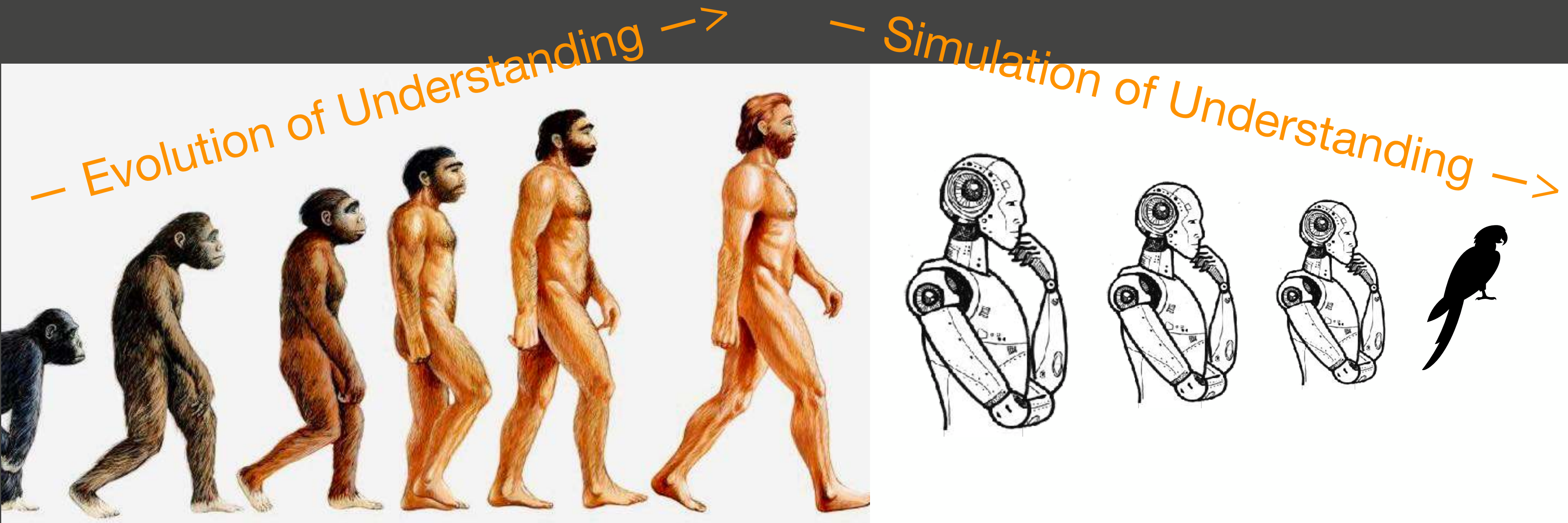
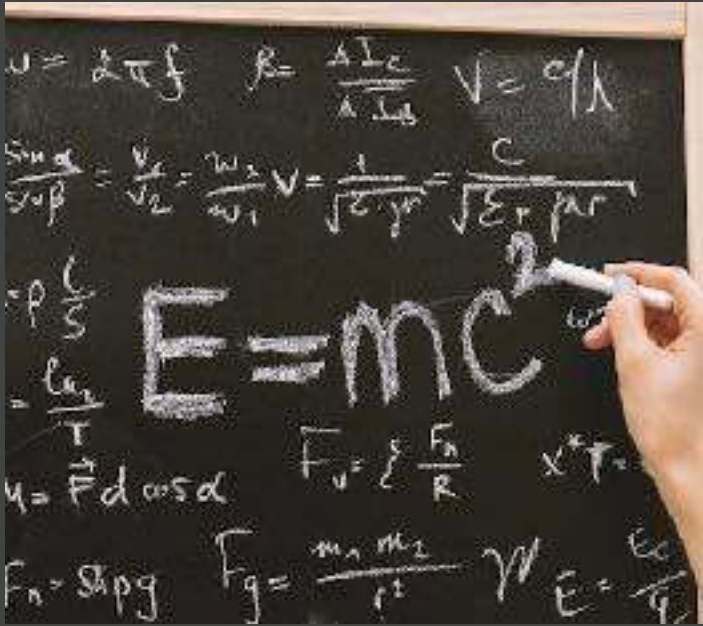
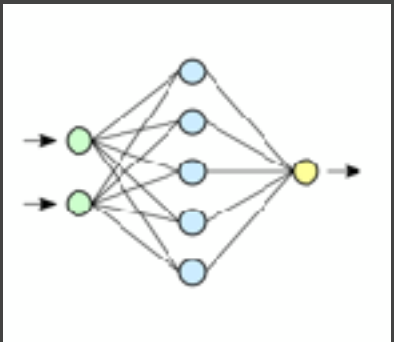
CURRENT AI: EVOLUTIONARY PROGRESS? REALLY?



1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100



1	2	3	4	5	6	7	8	9	10
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21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100



The prime-number parrot would likely be better than most humans — without having any clue about prime numbers itself

Source: <https://medium.com/buzzrobot/is-ai-the-future-of-evolution-fbe109602098>

Epistemology of science: data-driven AI may seem like a step backwards!

WHAT IS AI?



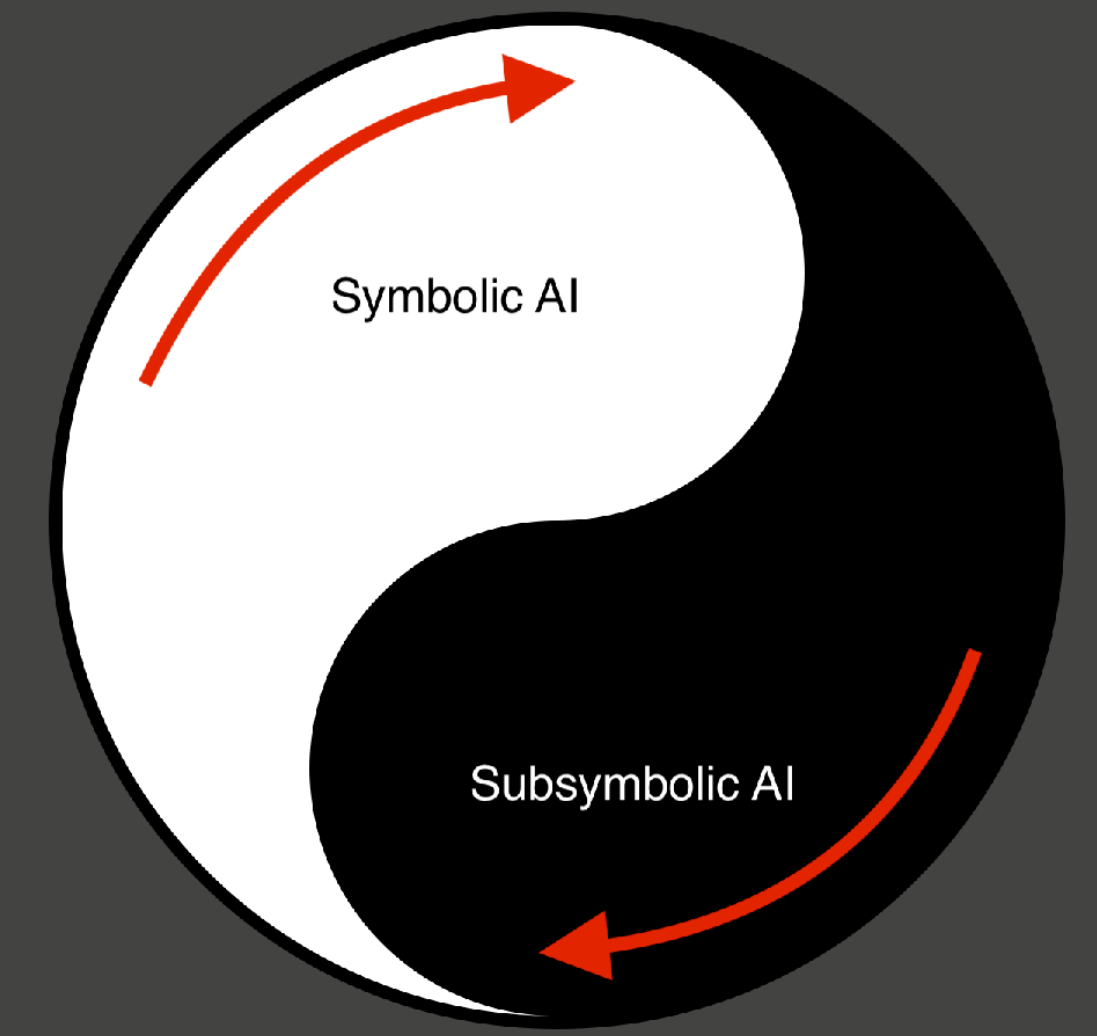
The School of Athens (detail) — by Raphael (courtesy of Wikimedia Commons)

Humans develop theories & arguments also from pure thought experiments
(*Example: Ontological Argument*)

WHAT IS AI?

TAKE AWAY POINTS

- **AI** has **many subfields** (including, but not limited to **ML**)
- There have been **successes** in **symbolic & subsymbolic AI**
- There have been **AI summers & AI winters** (next cooling period ahead?)
- AI is a **dual use technology** requiring adequate regulation
- **Data-driven ML** economically relevant — conceptually a **step backwards?**
- Next big thing: **hybrid AI** or **neuro-symbolic AI**



WHAT IS AI? SOME READING

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